

DIY: Hovercraft

On the brink of flight!

A DIY that will get you started on the first steps to creating your own flying machine

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Presenting Alok Deherkar, this intrepid young man spends almost all his time mucking about in his own little DIY accessories shop. Since the time he started his shop, Alok has designed innumerable gadgets and devices for everything from college projects to car and bike accessories. In his own words, "I'm a DIY for life". His ultimate goal is to create his own personal, home-made helicopter before he dies!

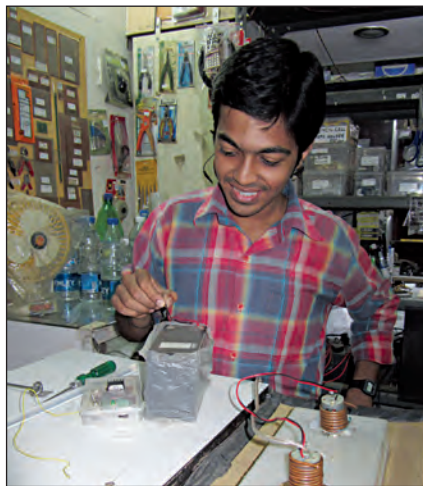
He designed his first hovercraft in August and has spent a lot of time in refining the design to its current state. We present you with the basic hovercraft design that he started off with. Remember, there are no hard and fast rules for designing this and you can spend a lot of time modifying and tweaking the design to your heart's content.

The motor housing

For the hovercraft, you can use any number of motors you choose. Alok used 2 for his design and they were more than sufficient. The motors he used are 11.1V, 18,000rpm motors (zero load) which he managed to salvage from a couple

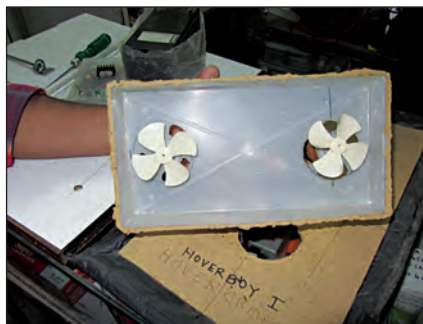


Getting started



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of blow-driers. The mounting point for the motors is an old plastic box.



Attaching the motors

On the bottom of the box, you need to carefully draw a line dividing the box exactly in half, lengthwise. Decide where you want to mount the motors and make sure that they are aligned and along the line you marked. It would be best if you balance the motors on the box itself by keeping them equidistant from the edges as this will save you a lot of hassle later when mounting the skirt.

For each motor you plan to use, cut a hole for the shaft along the central line

you

STUFF YOU'LL NEED

Cardboard; Garbage bag; Plastic box of some sort; At least two, high rpm motors; Power source (Battery pack whose voltage depends on the motors); At least two propellers; A precise ruler; Marker pen; Glue gun or glue of some sort; Long wires; A sharp blade, scissors, nails, etc.; Drill with fine bits (for motor and screw mounts).

marked earlier. Drill some screw holes to mount the motors and some vents for airflow, as shown in the images. Don't make the vents too large or else the structure will become weak and won't be able to support the weight and torque of the motors.

The copper coloured stuff you see wrapped around the motors, is, well, Copper. This tubing was added to the motor to aid in cooling as it gets hot after extended use and is strictly optional.



Cutting the cardboard

Attach the motors to the box by screwing them into place. Next, attach the propellers to the motor. If the shaft diameter is too large for the propellers,